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ORIGINAL RESEARCH ARTICLE

Effects of Autograft Types on Muscle Strength and Functional Capacity in Patients Having Anterior Cruciate Ligament Reconstruction: A Randomized Controlled Trial

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Abstract

Background The effects of different autograft types for anterior cruciate ligament reconstruction (ACL-R) on muscle function are sparsely investigated in randomized controlled trials. Our aim was to investigate the effects of quadriceps tendon autograft (QTB) vs. semitendinosus-gracilis autograft (StG) on thigh muscle strength and functional capacity, and a patient-reported outcome 1 year after ACL-R, and to compare the results to healthy controls.

Methods ACL-R patients ($n = 100$) and matched controls (CON, $n = 50$) were recruited, with patients being randomly assigned to QTB ($n = 50$) or StG ($n = 50$) ACL-R. One year after ACL-R, bilateral knee extensor (KE) and flexor (KF) muscle strength (isometric, dynamic, explosive, limb symmetry index [LSI], hamstring:quadriceps ratio [HQ ratio]) were assessed by isokinetic dynamometry, along with functional capacity (single leg hop distance [SHD]) and a patient-reported outcome (International Knee Documentation Committee [IKDC] score).

Results KE muscle strength of the operated leg was lower (9–11%) in QTB vs. StG as was KE LSI, while KF muscle strength was lower (12–17%) in StG vs. QTB as was KF LSI. HQ ratios were lower in StG vs. QTB. Compared with the controls, KE and KF muscle strength were lower in StG (10–22%), while KE muscle strength only was lower in QTB (16–25%). Muscle strength in the StG, QTB, and CON groups was identical in the non-operated leg. While SHD and IKDC did not differ between StG and QTB, SHD in both StG and QTB was lower than CON. The IKDC scores improved significantly 1 year following ACL-R for both graft types.

Conclusion One year after ACL-R, muscle strength is affected by autograft type, with StG leading to impairments of KE and KF muscle strength, while QTB results in more pronounced impairments of KE only. Functional capacity and patient-reported outcome were unaffected by autograft type, with the former showing impairment compared to healthy controls.

Clinical Trials Registration Number NCT02173483.

1 Introduction

Anterior cruciate ligament (ACL) injury is the most common ligamentous knee injury [1, 2] that frequently occurs during sports, such as soccer, handball, and alpine skiing [3, 4]. Following an ACL injury, treatment aims are for the patient to return to sports and symptom-free activities of daily life. If patients intend to return to pivoting sports activities (e.g., soccer and handball), ACL reconstruction (ACL-R) is often recommended [5].

Several measures of muscle strength have been applied when trying to determine the recovery level after ACL-R, but there is no consensus on what strength outcome to use in the literature. While peak torque of the knee extensor (KE) and knee flexor (KF) has been used most often, explosive muscle strength (rate of force development [RFD]), between-leg asymmetry known as the limb symmetry index (LSI; the difference between operated and non-operated legs) [6], and within-leg asymmetry known as the hamstring:quadriceps ratio (HQ ratio) [7] also seem important for understanding muscle function following ACL-R. At the same time, a comparison to

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Key Points

Anterior cruciate ligament reconstruction (ACL-R) using semitendinosus-gracilis autograft (StG) leads to impairments of both knee flexor and knee extensor muscle strength, whereas quadriceps tendon autograft (QTB) leads to pronounced impairments in knee extensor muscle strength only.

Functional capacity and patient-reported outcome appear to be unaffected by autograft type when undergoing ACL-R.

matched healthy controls would help elucidate whether the muscle function of ACL patients is influenced by a postoperative change of behavior (physical inactivity) and/or mindset (fear avoidance). Interestingly, no studies have thus far included a wide array of outcomes describing muscle strength as well as normative healthy data in a randomized controlled trial (RCT) to evaluate the effects of different autograft types following ACL-R.

In Denmark, the semitendinosus-gracilis autograft (StG) is preferred over the quadriceps tendon autograft (QTB; 85% vs. 8%, respectively) [8]. A couple of non-randomized studies have directly compared muscle strength in StG to QTB [9–14], with divergent results and large differences in both methods and study populations. Therefore, the effects of autograft type on a broad array of muscle strength outcomes following ACL-R is not fully clear; thus, an evaluation of these in a head-to-head RCT design in the general ACL population is required.

The purpose of this study was, therefore, to compare KE and KF muscle strength (peak torque, RFD, within- and between-leg asymmetry), functional capacity, and patient-reported outcomes in patients with either StG or QTB ACL-R 1 year postoperatively using an RCT design. A secondary aim was to compare ACL patients to a non-injured matched control group. It was hypothesized that: (1) when comparing patients with StG to QTB 1 year after ACL-R, StG patients would have impaired KF muscle strength, while QTB patients would have impaired KE muscle strength and (2) that both the operated and the non-operated leg of ACL patients would, independent of graft type, show reduced muscle strength, when compared to a matched healthy control group.

2 Methods

2.1 Study Design and Approvals

A prospective randomized controlled clinical trial was conducted with assessments performed 1 year after ACL-R (Table 1). The study was registered at http://www.clinicaltrials.gov (NCT02173483), approved by the local scientific ethical committee (Journal no. 110928514) and by the Danish Data Protection Agency, and conducted in accordance with the Helsinki Declaration. All participants provided written consent. The outcome of interest to the present study was maximal voluntary muscle strength (a secondary outcome in the overall study). The sample size calculation was nevertheless determined from the primary overall study outcome, the International Knee Documentation Committee (IKDC) score (minimal relevant delta change = 7 points,

Table 1 Participant characteristics 1 year after ACL-R

	StG ($n = 43$)	QTB ($n = 42$)	CON ($n = 50$)	Group effect p values
Age (years)	28.3 (6.2)	28.7 (6.4)	28.3 (6.2)	= 0.87
Weight (kg)	75.6 (13.0)	77.0 (11.5)	75.4 (15.7)	= 0.71
Height (cm)	176 (8.6)	175 (8.9)	176 (11.2)	= 0.82
Body Mass Index (kg/m^2)	24.3 (3.2)	25.1 (3.2)	24.0 (3.1)	= 0.26
Male:female ratio, n (%)	23/20 (53/47)	25/17 (60/40)	27/23 (54/46)	= 0.78
Injury to ACL-R (months)	14.6 (15.1)	14.4 (20.5)	–	= 0.43
Time since ACL-R (months)	12.5 (0.7)	12.4 (0.6)	–	= 0.74
Meniscus injury (n)	18	16	–	= 0.72
Meniscus repair/resection (n)	4/13	7/8	–	= 0.72
Meniscus injury without treatment (n)	1	1	–	= 0.99
Cartilage injury ICRS > 2 (n)	1	3	–	= 0.29
Perioperative complications[a] (n)	0	0	–	n.c.

Data are shown as mean ± SD except for male:female ratio shown as n and percentage

n.c. not computable

[a]Deep infections, hardware removal, secondary surgery, deep venous thrombosis

standard deviation ≈ 12 points, β = 0.80, α = 0.05), as this is a well-established clinical outcome contrary to muscle strength in ACL patients. When accounting for variability in IKDC along with an expected participant attrition, the sample size for each group was $n = 50$ (from a calculated $n = 37$).

A separate paper has addressed numerous patient-reported outcomes 1 and 2 years after ACL-R [15].

2.2 Study Population

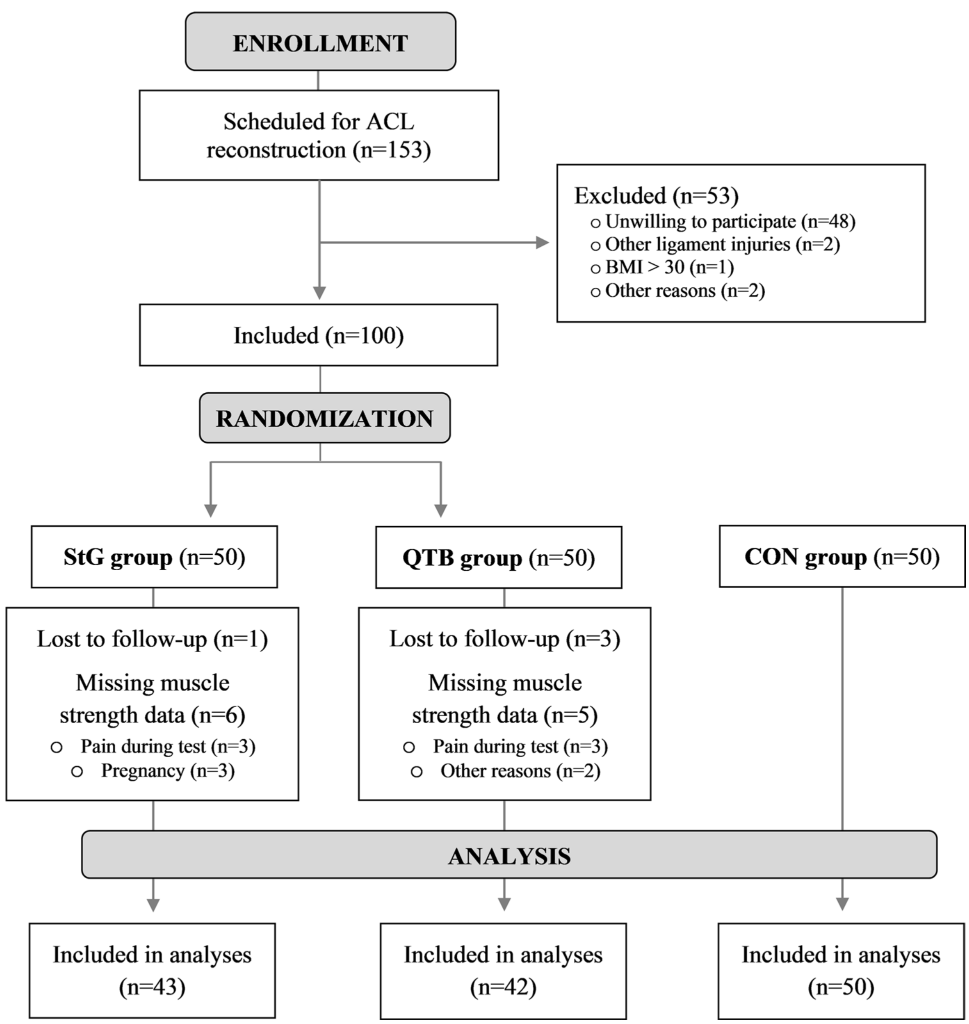
All patients over the age of 18 years who were scheduled for isolated ACL-R at Aarhus University Hospital orthopedic clinic between May 2014 and–January 2016 were invited to participate in the study. Exclusion criteria were diabetes, arthritis, active malignant disease, other ligamentous instability, Morbus Bechterew (ankylosing spondylitis), other disabling musculoskeletal illness, BMI > 30, current treatment with glucocorticoid drugs or growth hormone, and expected inability to complete the standard rehabilitation program. No patients were excluded because of meniscus or cartilage pathology. All ACL-R patients followed a standard rehabilitation regimen which is described below. Figure 1 presents the study flowchart.

Healthy controls (CON) were recruited by advertising in different locations on campus, on Facebook, and from the friends, colleagues, and relatives of the ACL participants and authors. The healthy controls were matched for gender and age (± 5 years) in a 1:2 ratio to the ACL_R patients. Exclusion criteria for CON were the same as for patients (self-reporting), except that previous treatment for knee injury was added to the criteria.

2.3 Randomization

The participants were randomized using concealed numbered envelopes containing either QTB- or StG-marked pieces of paper. The surgeon opened the envelope just prior to surgery and informed the patient about the graft type after the surgery. As the surgery induced scar that were procedure specific, blinding was not possible for surgeons, patients or evaluators.

Fig. 1 Flowchart of participants



2.4 Surgery

Six experienced surgeons performed all ACL-R surgeries arthroscopically. StG ACL-R was performed with four-strand gracilis and semitendinosus grafts fixed at the femur with Endobutton CL (Smith & Nephew, Andover, USA) and in the tibia with a PEEK interference screw (Smith & Nephew, Andover, USA). QTB ACL-R were performed with 5 mm partial-thickness grafts with 70 mm long soft tissue grafts and an additional 20 mm long patella bone block. Femoral fixation was performed with a titanium interference screw (Arthrex, Naples, USA), and in the tibia, with a PEEK interference screw (Smith & Nephew, Andover, USA). Hamstring grafts were harvested through an oblique incision over the pes anserinus, and quadriceps tendon graft was harvested through a 4 cm longitudinal incision over the tendon insertion at the patella. The femoral drill hole position was placed at the anteromedial part of the ACL insertion area at the femoral condyle by visualizing and drilling inside-out from an anteromedial portal. The tibia tunnel was drilled outside-in with an aiming guide centered at the stump of the ACL tibial insertion in line with the interior margin of the lateral meniscus forehorn. Overall, graft diameter varied from 7.5 to 9.0 mm.

2.5 Postoperative Rehabilitation

The standard rehabilitation program was based on the following progressive approach: days 1–14: full support to pain threshold, free movement, no bandages; weeks 3–12: frequent movement exercises supervised by a physiotherapist (patients were offered 1 weekly training session during the first 3 months), bicycle ergometer, full weight bearing; months 4–9: running allowed; months 10–12: contact sports allowed. While patients who underwent meniscus repair or resection received the same overall rehabilitation program as those who did not, minor individual adjustments did take place.

2.6 Muscle Strength and Muscle Strength Asymmetry

Muscle strength of the KE and KF was tested using an isokinetic dynamometer (Humac Norm, CSMI, Stroughton, USA), which is considered the "gold-standard" assessment, [16, 17] as described in detail elsewhere [18, 19]. Both the non-operated and operated legs were tested in ACL-R patients (termed QTB_{con}, QTB_{op}, StG_{con}, and StG_{op}, respectively), with the non-operated leg always being the starting leg. While the testing order for the legs of CON was determined by randomization, the starting leg was designated as the "non-operated leg" and vice versa ($CON_{'con'}$ and $CON_{'op'}$) to match the assessment procedure in ACL-R patients.

After a 5-min warm-up at moderate intensity on a stationary bike (Monark 818 Ergomedic, Vansbro, Sweden), muscle strength assessments were performed in the following order: 0 deg/s isometric, 60 deg/s and 180 deg/s concentric, and − 60 deg/s eccentric, and always with KE preceding KF measurements. Knee joint angles during isometric tests were 70 and 20 deg (full knee extension = 0 deg) for KE and KF, respectively. The range of motion during dynamic tests were between 5 and 100. Subjects were given one familiarization trial before performing 3–5 maximal trials at all contraction types. Subjects were instructed to perform the trials as fast and as powerfully as possible while receiving verbal encouragement. All muscle strength data was sampled at 1500 Hz and subsequently analyzed using custom-made software (LabVIEW 2011, National Instruments Corporation, TX, USA). Muscle strength was determined from the subject's best trial for each contraction type. Explosive muscle strength (rate of force development = Δtorque/Δtime, RFD) was derived from isometric trials during the early (0–50 ms) and late phases (0–200 ms) of muscle contractions. To allow for a direct comparison between the groups, muscle strength data was normalized to body weight (Nm/kg) as were RFD (Nm/s/kg). Testing was performed by two of the authors (KS and TN), but without blinding of graft type due to the visibility of the operation scars.

Muscle strength asymmetry was assessed between the operated and non-operated legs using LSI (= operated leg/non-operated leg × 100), and between KE and KF muscle groups (within leg) using the HQ ratio (= hamstring/quadriceps muscle strength) [20, 21]. Different HQ ratios were calculated according to the conventional approach (KF/KE at − 60, 0, 60, and 180 deg/s, respectively) and to the functional approach (knee extension: KF − 60 deg/s/KE 60 deg/s; knee flexion: KF 60 deg/s/KE − 60 deg/s). The functional approach is considered to be more strongly associated with functional capacity [22].

2.7 Functional Capacity

As an indicator of functional capacity, participants performed the single leg hop distance (SHD), which involved standing on one leg, jumping as far as possible, and landing on the same leg (distance measured from starting-to-landing toe position). Both hands had to be placed behind the back during the entire test. For a trial to be accepted, the participant had to perform a controlled landing and had to hold the foot/landing position for > 2 s [18, 23, 24]. Participants were given three practice trials followed by three full effort trials. ACL-R patients always started with the non-operated leg (termed "non-operated" leg for CON). SHD is considered a reliable outcome measure in ACL patients [24–26].

2.8 Patient-Reported Outcome

As a measure of patient-reported outcome, the IKDC score [27, 28] was assessed for the operated leg in all ACL patients immediately prior to ACL-R and at the 1-year follow-up.

2.9 Statistical Analysis

Analyses were performed using a linear mixed model (STATA 14.2, StataCorp, USA). Data were tested for normal distribution, and if absent, appropriate transformations were carried out (all HQ-ratio data were log transformed). Measurements of muscle strength and functional capacity were analyzed with Group (QTB, StG, CON) as a fixed effect and Participant ID as a random effect. Whenever a between-group difference was present, post hoc analyses were carried out to reveal specific group differences. As dominant leg, age, and gender could potentially affect the outcomes, secondary analyses adjusting for these confounders, were also undertaken. Measurements of patient-reported outcome (IKDC) were analyzed with Time (prior to ACL-R and at the 1-year follow-up) and Group (QTB, StG) as a fixed effect and Participant ID as a random effect. Data are presented as mean ± SD in tables and as mean ± 95% CI in figures. The level of statistical significance was set at $p<0.05$, along with trends towards statistical significance set at $p<0.10$. For comparisons of CON with StG or QTB, data are marked with bold in tables and grey symbols in figures. All statistical analyses were performed by an assessor blinded for groups.

3 Results

3.1 Participant Characteristics

As shown in Table 1, the participants' characteristics did not differ between any of the groups 1 year after ACL-R.

3.2 Effect of Autograft on Muscle Strength and Muscle Strength Asymmetry

An overall effect of group was observed for all muscle strength outcomes (eccentric, isometric, dynamic, RFD) and asymmetry outcomes (LSI, HQ ratio) in the operated leg (Fig. 2a–d, Tables 2 and 3), except for KF RFD_{50ms} LSI (Table 3). In the non-operated leg, no overall effect of group was observed (Fig. 2f, Tables 2 and 3).

When comparing the two autograft patient groups, 4%–11% lower muscle strength was observed during KE at 0 deg/s (isometric, MVIC) and 60 deg/s (concentric dynamic) in QTB vs. StG, along with 4% lower KE RFD_{200ms} (Fig. 2a, Table 2). In contrast, 13–17% lower muscle strength was observed during KF at 0 deg/s (isometric, MVIC), 60 deg/s, and 180 deg/s (concentric dynamic) in StG vs. QTB, along with 20% and 21% lower KF RFD_{50ms} and RFD_{200ms} values (Fig. 2b, Table 2). Lower LSI values were observed in QTB vs. StG across all angular velocities and for both RFD phases during KE, whereas lower LSI values were observed in StG vs. QTB across all angular velocities and RFD_{200ms} (but not RFD_{50ms}) during KF (Table 3). In addition, all HQ-ratio values were lower in StG vs. QTB (Table 3).

In patients having StG autograft vs. CON, 10–22% lower muscle strength was observed across all angular velocities and RFD phases during both KE and KF in the operated leg (Fig. 2a, b, Table 2). In addition, all LSI values were lower in StG vs. CON during both KE (except for RFD_{50ms}) and KF (Fig. 2c, d, Table 3), whereas no differences were observed for any of the HQ-ratio values (Table 3).

For patients with QTB autografts vs. CON, 16–25% lower muscle strength was observed across all angular velocities and RFD phases during KE but not during KF in the operated leg (Fig. 2a, b, Table 2). Also, all LSI values were lower in QTB vs. CON during KE but not during KF (Fig. 2c, d, Table 3), and all HQ-ratio values were higher in QTB vs. CON (Table 3).

Adjusting for age, gender, and dominant leg did not influence any of the findings described above.

3.3 Effect of Autograft on Functional Capacity

An overall effect of group was observed for SHD in both the operated leg (Fig. 2e, Table 2) and the non-operated control leg (Table 3). Furthermore, an overall group effect was observed for SHD LSI (Table 3).

As evidenced from Fig. 2e and Table 2, no SHD difference was found between StG and QTB for the operated leg, whereas SHD was 17% and 19% lower in StG and QTB vs. CON, respectively, in the operated leg. A similar pattern was apparent in the non-operated control leg, with SHD 12% and 10% lower in StG and QTB vs. CON, respectively (Table 2). For the SHD LSI, StG and QTB did not differ, while SHD LSI was lower in StG vs. CON and in QTB vs. CON (Table 3).

Adjusting for age, gender, and dominant leg did not influence any of the findings described above.

3.4 Effect of Autograft on Patient-Reported Outcome

While no IKDC score differences were found between StG and QTB graft groups prior to ACL-R or at the 1-year follow-up, significant improvements at the 1-year timepoint were observed in both groups with StG increasing from 60 ± 14 to

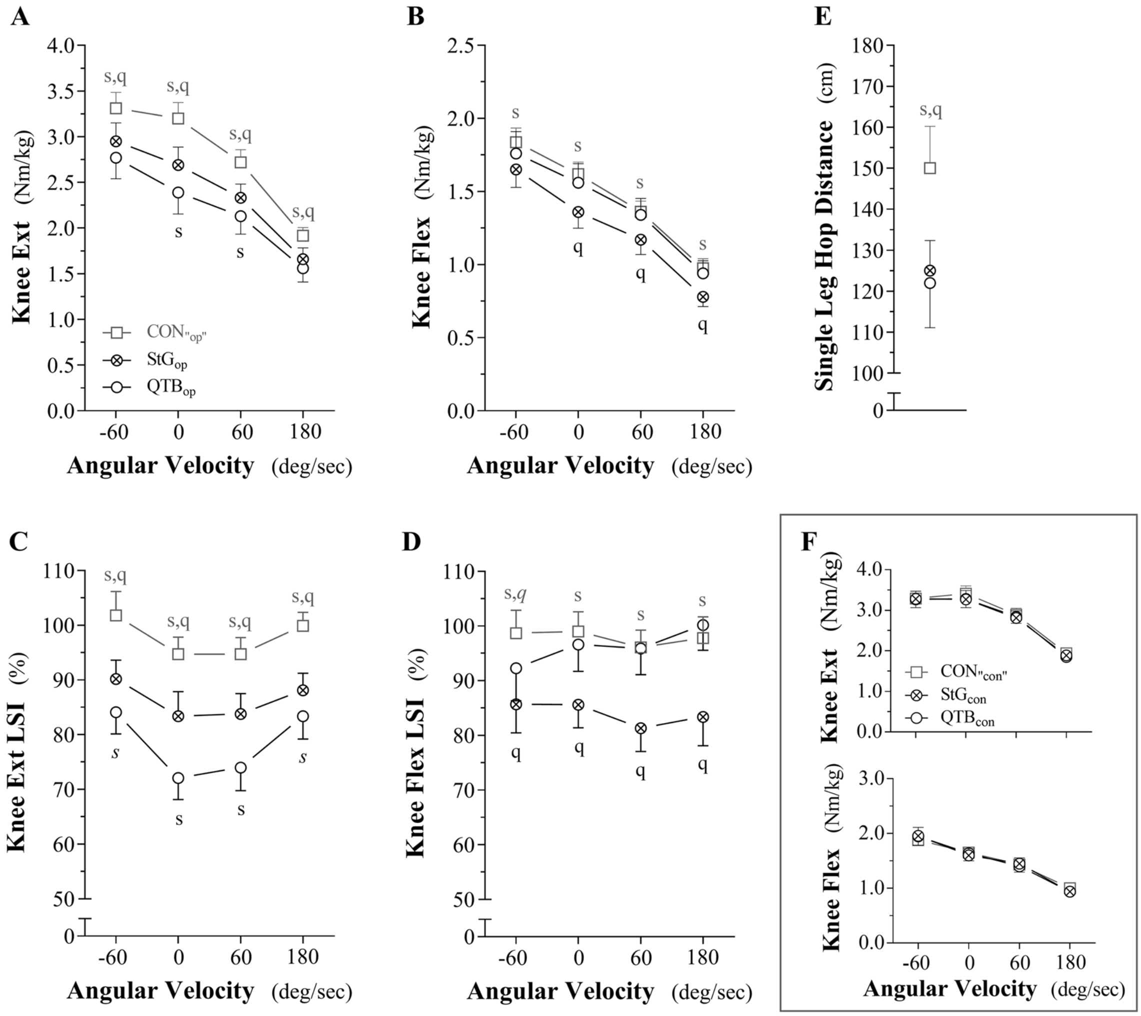


Fig. 2 Muscle strength across different angular velocities (and contraction types) and limb symmetry index (LSI) during knee extension (**a**, **c**) and knee flexion (**b**, **d**), along with single leg hop distance (SHD; E), are displayed for the "operated leg" of CON (control group, squares), StG (semitendinosus-gracilis autograft group, crossed circles), and QTB (quadriceps tendon bone plug autograft, open circles). Also, muscle strength across different angular velocities during knee extension (KE) and knee flexion (KF) are displayed for the non-operated control leg (**f**). Data are shown as mean ± 95% CI. Symbols s and q indicate significant difference ($p < 0.05$) from StG or QTB, respectively (symbols in italic indicate trend, $p < 0.10$)

76 ± 15, $p < 0.05$ and QTB increasing from 56 ± 18 to 76 ± 17, $p < 0.05$.

4 Discussion

4.1 Main Findings

The main findings of the present study were that 1 year after ACL-R, patients with a StG autograft showed equally impaired muscle strength of both the KE and KF compared to CON, while patients with a QTB autograft only showed impairment of the KE muscle strength when compared to CON. These impairments of the KE muscle strength were nevertheless more pronounced in QTB than in StG, as also evidenced by lower KE muscle strength in QTB compared to StG. Muscle strength of the contralateral leg did not differ between StG, QTB or CON. Furthermore, impaired functional capacity and patient-reported outcome were evident for the operated leg in both ACL groups compared to CON (yet with no differences between StG and QTB), along with impaired functional capacity in the non-operated leg compared to CON (but to a lesser degree than in the operated leg). Also, the patient-reported subjective outcome (IKDC score) improved at the 1-year follow-up after ACL-R. A more detailed investigation of the clinical patient-reported outcomes has been published in a separate paper [15].

Table 2 Peak torque across different contraction velocities (and contraction types), RFD, and SHD in StG, QTB, and CON 1 year after ACL-R

Operated leg		StG_{op}	QTB_{op}	$CON_{“op”}$	Group effect p values	Posthoc analysis
KE − 60°/s	Nm/kg	2.95 ± 0.66	2.77 ± 0.74	3.31 ± 0.62	< 0.01	**StG < CON, QTB < CON**
KE 0°/s	Nm/kg	2.69 ± 0.64	2.39 ± 0.76	3.20 ± 0.62	< 0.01	**StG < CON, QTB < CON**, QTB < StG
KE 60°/s	Nm/kg	2.33 ± 0.49	2.13 ± 0.63	2.72 ± 0.49	< 0.01	**StG < CON, QTB < CON**, QTB < StG
KE 180°/s	Nm/kg	1.66 ± 0.40	1.56 ± 0.48	1.92 ± 0.32	< 0.01	**StG < CON, QTB < CON**
KE RFD_{50ms}	Nm/s/kg	23.0 ± 11.7	22.2 ± 8.7	26.7 ± 8.8	= 0.03	**StG < CON(.08), QTB < CON**
KE RFD_{200ms}	Nm/s/kg	10.2 ± 2.7	9.8 ± 4.0	12.3 ± 2.6	< 0.01	**StG < CON, QTB < CON**, QTB < StG
KF − 60°/s	Nm/kg	1.65 ± 0.40	1.76 ± 0.48	1.84 ± 0.34	= 0.07	**StG < CON**
KF 0°/s	Nm/kg	1.36 ± 0.36	1.56 ± 0.42	1.62 ± 0.29	< 0.01	**StG < CON**, StG < QTB
KF 60°/s	Nm/kg	1.17 ± 0.33	1.34 ± 0.36	1.36 ± 0.26	< 0.01	**StG < CON**, StG < QTB
KF 180°/s	Nm/kg	0.78 ± 0.22	0.94 ± 0.28	0.97 ± 0.23	< 0.01	**StG < CON**, StG < QTB
KF RFD_{50ms}	Nm/s/kg	9.9 ± 3.7	12.3 ± 6.1	12.7 ± 5.1	< 0.01	**StG < CON**, StG < QTB
KF RFD_{200ms}	Nm/s/kg	5.4 ± 1.6	6.8 ± 2.8	6.7 ± 1.2	< 0.01	**StG < CON**, StG < QTB
SHD	Cm	125 ± 24	122 ± 35	150 ± 36	< 0.01	**StG < CON, QTB < CON**
Non-operated leg		StG_{con}	QTB_{con}	$CON_{“con”}$	Group effect	Posthoc analysis
KE − 60°/s	Nm/kg	3.28 ± 0.62	3.27 ± 0.66	3.29 ± 0.63	= 0.99	
KE 0°/s	Nm/kg	3.27 ± 0.72	3.28 ± 0.70	3.41 ± 0.69	= 0.55	
KE 60°/s	Nm/kg	2.81 ± 0.58	2.84 ± 0.53	2.89 ± 0.56	= 0.75	
KE 180°/s	Nm/kg	1.89 ± 0.40	1.85 ± 0.38	1.94 ± 0.39	= 0.52	
KE RFD_{50ms}	Nm/s/kg	29.8 ± 11.1	28.0 ± 10.1	30.3 ± 9.4	= 0.56	
KE RFD_{200ms}	Nm/s/kg	13.7 ± 3.8	13.0 ± 3.2	14.2 ± 2.9	= 0.28	
KF − 60°/s	Nm/kg	1.96 ± 0.50	1.95 ± 0.52	1.88 ± 0.32	= 0.64	
KF 0°/s	Nm/kg	1.60 ± 0.41	1.64 ± 0.44	1.65 ± 0.31	= 0.80	
KF 60°/s	Nm/kg	1.45 ± 0.34	1.40 ± 0.34	1.43 ± 0.30	= 0.78	
KF 180°/s	Nm/kg	0.94 ± 0.20	0.94 ± 0.28	1.00 ± 0.22	= 0.33	
KF RFD_{50ms}	Nm/s/kg	11.5 ± 4.8	11.7 ± 4.5	12.1 ± 3.6	= 0.84	
KF RFD_{200ms}	Nm/s/kg	6.6 ± 1.8	6.7 ± 2.1	6.9 ± 1.5	= 0.67	
SHD	Cm	129 ± 23	132 ± 31	147 ± 33	< 0.01	**StG < CON, QTB < CON**

Data are presented as mean ± standard deviation. Post hoc analyses between QTB and CON as well as between StG and CON are shown in bold (trends shown with p value in brackets)

KE knee extensors, *KF* knee flexors, *SHD* single leg hop distance

4.2 Comparison to Other Literature

Non-randomized studies have predominantly compared muscle strength in patients having either QTB or StG. A study by Fischer et al. [12] compared muscle strength in StG vs. QTB 6–8 months after ACL-R. Similar to our findings, a lower KE muscle strength and a higher HQ ratio were observed in QTB, but no significant difference in KF muscle strength was found. Also in concordance with our study, Fischer et al. reported higher LSI KE scores in StG patients, while QTB patients had higher scores for LSI KF at both test points. A recent cohort study by Kuenze et al. [9] reported findings comparable to Fischer et al. [12] (i.e., lower KE muscle strength and LSI in QTB vs. StG 7 months after ACL-R). However, in ACL patients examined 30 months after ACL-R, no differences were observed between QTB and StG in KE muscle strength or LSI [9]. The minor discrepancies between studies might relate to different timepoints of postoperative testing and the lack of randomization. Lee et al. [13] compared StG and QTB in a cohort study 1 and 2 years postoperatively. At both the 1- and 2-year timepoints, they found results comparable to the present study regarding StG having reduced KF strength compared to QTB, but they found no significant difference in KE strength. No clear methodological differences explaining the divergent findings in the present study could be identified. Martin-Alguacil

Table 3 LSI and HQ ratio in StG, QTB, and CON 1 year after ACL-R

LSI		$StG_{op/con}$	$QTB_{op/con}$	$CON_{\text{“op/con”}}$	Group effect *p* values	Posthoc analysis
KE − 60°/s	%	90.2 ± 11.2	84.1 ± 12.7	101.8 ± 15.4	< 0.01	**StG < CON, QTB < CON**, QTB < StG
KE 0°/s	%	83.4 ± 14.6	72.1 ± 12.7	94.7 ± 11.0	< 0.01	**StG < CON, QTB < CON**, QTB < StG
KE 60°/s	%	83.8 ± 12.2	74.0 ± 13.6	94.7 ± 10.7	< 0.01	**StG < CON, QTB < CON**, QTB < StG
KE 180°/s	%	88.1 ± 10.2	83.4 ± 13.4	99.9 ± 8.7	< 0.01	**StG < CON, QTB < CON**, QTB < StG(.06)
KE RFD_{50ms}	%	81.7 ± 27.8	78.5 ± 26.5	90.6 ± 21.2	= 0.09	**QTB < CON**
KE RFD_{200ms}	%	78.3 ± 17.7	73.2 ± 18.5	87.7 ± 12.8	< 0.01	**StG < CON, QTB < CON**, QTB < StG
KF − 60°/s	%	85.7 ± 17.0	92.3 ± 19.2	98.7 ± 14.8	< 0.01	**StG < CON, QTB < CON(.07)**, StG < QTB
KF 0°/s	%	85.6 ± 13.7	96.6 ± 15.8	99.0 ± 12.7	< 0.01	**StG < CON**, StG < QTB
KF 60°/s	%	81.3 ± 13.8	95.9 ± 15.4	96.1 ± 11.2	< 0.01	**StG < CON**, StG < QTB
KF 180°/s	%	83.4 ± 17.2	100.2 ± 14.9	97.8 ± 13.7	< 0.01	**StG < CON**, StG < QTB
KF RFD_{50ms}	%	95.3 ± 45.4	101.2 ± 32.3	116.3 ± 56.0	= 0.16	
KF RFD_{200ms}	%	84.0 ± 43.0	97.4 ± 53.8	101.9 ± 33.0	< 0.01	**StG < CON**, StG < QTB
SHD	%	97.1 ± 11.3	92.1 ± 16.1	104.0 ± 18.5	< 0.01	**StG < CON, QTB < CON**
HQ $ratio_{operated\ leg}$		StG	QTB	CON	Group effect	Posthoc analysis
KF:KE − 60°/s		0.57 ± 0.12	0.65 ± 0.13	0.57 ± 0.11	< 0.01	**QTB > CON**, StG < QTB
KF:KE 0°/s		0.51 ± 0.10	0.68 ± 0.16	0.51 ± 0.07	< 0.01	**QTB > CON**, StG < QTB
KF:KE 60°/s		0.51 ± 0.11	0.64 ± 0.13	0.51 ± 0.07	< 0.01	**QTB > CON**, StG < QTB
KF:KE 180°/s		0.72 ± 0.18	0.88 ± 0.18	0.71 ± 0.09	< 0.01	**QTB > CON**, StG < QTB
KF − 60°/s:KE 60°/s		0.71 ± 0.15	0.86 ± 0.19	0.68 ± 0.12	< 0.01	**QTB > CON**, StG < QTB
KF 60°/s:KE − 60°/s		0.40 ± 0.11	0.49 ± 0.11	0.42 ± 0.08	< 0.01	**QTB > CON**, StG < QTB
HQ $ratio_{non\text{-}operated\ leg}$		StG	QTB	CON	Group effect	Posthoc analysis
KF:KE − 60°/s		0.60 ± 0.11	0.59 ± 0.11	0.58 ± 0.09	= 0.60	
KF:KE 0°/s		0.49 ± 0.08	0.50 ± 0.10	0.49 ± 0.07	= 0.77	
KF:KE 60°/s		0.52 ± 0.10	0.49 ± 0.08	0.50 ± 0.07	= 0.22	
KF:KE 180°/s		0.78 ± 0.16	0.76 ± 0.12	0.74 ± 0.09	= 0.37	
KF − 60°/s:KE 60°/s		0.70 ± 0.13	0.69 ± 0.15	0.66 ± 0.10	= 0.19	
KF 60°/s:KE − 60°/s		0.45 ± 0.10	0.43 ± 0.09	0.44 ± 0.09	= 0.67	

Data are presented as mean ± standard deviation. Post hoc analyses between QTB and CON as well as between StG and CON are shown in bold (trends shown with *p* value in brackets)

KE knee extensors, *KF* knee flexors, *SHD* single leg hop distance

et al. [29] reported higher HQ ratios in QTB vs. StG 1 year after ACL-R but comparable functional capacity, which was also found in the present study. However, the level of the HQ ratios was much lower in the present study (~ 0.6 vs. ~ 1.1), reflecting that KE was higher than KF in contrast to the study by Martin-Alguacil, where KF was often equal to or higher than KE. This may be explained by differences in the calculations of the HQ ratios (not explained by Martin-Alguacil et al.) or by differences in patient populations (all patients were soccer players in the study by Martin-Alguacil et al.). Recently, Huber et al. [10] compared muscle strength prior to ACL-R and 5 and 9 months following ACL-R with either QTB or StG in 464 patients. However, in their study, only 73 patients were examined prior to ACL-R, and 215 patients were examined 9 months following ACL-R. The only differences between QTB and StG were observed at the 5-month follow-up, where KE muscle strength and LSI were lower in QTB vs. StG, while lower KF muscle strength and LSI were seen in StG. In contrast, no differences in muscle strength were observed at the 9-month follow-up, thus differing from our findings. This discrepancy might be due to the fact that Huber et al.'s study lacked randomization and only isokinetic muscle strength at 120°/s was tested—a velocity not examined in the present study.

4.3 Clinical Implications

The present study raises some important clinical implications. Following ACL-R, it has previously been suggested that a return to sports should optimally be guided by an evaluation of both muscle strength and functional performance, rather than solely relying on the "time since surgery" criteria

[18]. In support, a recent study by Welling et al. found that only 21% passed the return-to-sport criteria of LSI > 90% when tested 9 months after ACL-R [30]. In our study, we similarly found that 22% and 29% of QTB and StG, respectively, passed the LSI > 90% when tested 12-month post ACL-R. Of note, only 71% of CON passed the criteria, raising some concern about the validity of the LSI criteria. Furthermore, impaired KF muscle strength increased the risk of further ACL injuries [31–33], while impaired KE muscle strength has been linked to changes in gait, functional performance, lower patient-reported outcome measurements, and an increased risk of osteoarthritis [34–36]. While we observed that graft choice had an impact on muscle strength 1 year after ACL-R, this was not the case for functional capacity and patient-reported outcome. The latter two thus appear less sensitive compared to muscle strength in capturing the effects of different graft types—information that could prove useful in designing post ACL-R rehabilitation programs tailored specifically toward the known consequences of a specific graft (i.e., a QTB graft would require interventions targeting the KE, while a StG graft would require interventions targeting both KE and KF).

Moreover, it appears that existing standard rehabilitation programs do not sufficiently re-establish muscle strength (as well as functional capacity and patient-reported outcome) 1 year following ACL-R to recommend a return to sports for the majority of patients. In that regard, one interpretation of the current data is that it will likely take longer to normalize KE muscle strength in persons having a QTB graft (displaying pronounced impairment in muscle strength compared to CON) than in persons having a StG graft (displaying moderate impairment in muscle strength compared to CON). Consequently, a more profound disturbance of between-leg (LSI) and within-leg (HQ ratio) asymmetry was observed in QTB, which will potentially increase the risk of injuries during the rehabilitation program and upon a return to sports. Taken together, this would point to the StG graft as the preferred graft choice if only the HQ ratio and LSI are considered. Altogether, the present findings appear to indicate that a more efficient standard rehabilitation program that accelerates recovery and/or longer periods of rehabilitation are warranted following ACL-R, independent of the choice of autograft [18]. However, the rehabilitation programs and compliance were not tracked for this study along with development of new injuries; thus, further investigation on these issues are warranted.

Interestingly, we found no muscle strength differences in the patients' contralateral legs when compared to CON. Therefore, the present study does not indicate a long-lasting postoperative change of behavior (physical inactivity) and/or mindset (fear avoidance) impacting muscle strength in the non-operated leg following ACL-R.

Patient-perceived outcome based on IKDC scores demonstrated improvements during the first year after ACL-R to an extent that surpassed what is the minimum clinically important improvement [28]. However, as mentioned above, no differences were seen between autograft types in the improvements of IKDC, fully in line with previous observations [9, 13].

Our study helps expand on the already established knowledge about postoperative muscle strength when choosing between different autograft types for anterior cruciate ligament reconstruction. However, it should be mentioned that selecting autograft type based solely on 1-year postoperative muscle strength data would be premature.

4.4 Limitations

A number of limitations should be considered when interpreting the present results. First, no data on the patients' rehabilitation programs were collected for this study; yet we assume that the RCT design of our study balanced inter-individual differences. Second, we did not assess the patients' muscle strength prior to ACL-R, although this has been shown feasible and useful when interpreting follow-up data [10]. Third, we only reported on a single patient-reported outcome here, with a separate paper addressing numerous patient-reported outcomes one and two after ACL-R [15].

5 Conclusion

At the 1-year timepoint after ACL-R, muscle strength, compared to the healthy controls, was affected by the type of autograft, with StG leading to moderate and quite similar impairments of KE and KF muscle strength, while QTB resulting in a more pronounced impairment of KE only, when compared to healthy controls. Functional capacity (single leg hop distance) and patient-reported outcome (IKDC score) were unaffected by graft type but impaired when compared to healthy controls.

Compliance with Ethical Standards

Ethics approval This study was performed in accordance with the ethical standards of the Helsinki Declaration. Ethics approval was obtained from the Regional Ethical Review Board (Central Denmark Region, Journal no. 110928514). The study was, furthermore, reported to the Danish Data Protection Agency.

Informed consent All participants provided informed consent.

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Conflict of interest The authors, Kasper Staghøj Sinding, Torsten Grønbech Nielsen, Lars Grøndahl Hvid, Martin Lind, and Ulrik Dalgas, declare that they have no conflicts of interest relevant to the content of this article.

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